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The performance of vertically perforated clay block masonry in fire tests predicted by a finite-element model including an energy-based criterion to identify spalling

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ABSTRACT

Fire tests on masonry are one of the most expensive experiments in developing new vertically perforated clay block geometries. Numerical simulations might be a reasonable substitute for such experiments, leading to a significant cost reduction in the development phase. However, the prediction of such tests with numerical modeling concepts is challenging due to large temperature and stress gradients, highly non-linear material effects, and the complex geometry of the blocks. Herein, we present a finite-element-based concept, including thermal and mechanical simulations, a unit-cell approach, a smeared damage model, and a novel energy-based spalling criterion to describe the structural and material behavior of a masonry wall in a fire experiment. We could predict the obtained spalling times of longitudinal webs and the total endurance of a masonry wall without any empirical fitting parameters in good agreement with experimental data. These results show that we can use our modeling approach for simulating such fire tests, enabling a much cheaper and more efficient development of block geometries.

1. Introduction

Fired clay block masonry obtains its strength and stiffness through a controlled firing process of the main constituents: the blocks. Nevertheless, fire poses a crucial impact on the load-carrying capacity of fired clay block masonry. Therefore, elaborate experiments are necessary to predict how resilient a masonry wall is to fire. These large-scale experiments are time-consuming, require special testing equipment, and, therefore, are costly. A review of different testing methods under fire was recently published by Daware and Naser [1].

The rapid advancements in computational methods and the increasing performance of modern computers allow the simulation of complex and highly non-linear problems in a reasonable amount of time. Hence, shifting the current procedure from these elaborate experiments to simulations seems obvious. However, these experiments are compulsory to obtain a certain level of product certification regarding fire safety according to EN 13501-2 [2]. Nonetheless, simulations seem to be a reasonable substitute for the experiments when comparing different block designs before bringing the best one to certification.

A growing group of scientists is trying to advance research considering the simulation of masonry under elevated temperatures. Kumar and Srivastava [3] published a comprehensive review of numerical models for structural frames with masonry infills in case of fire. Different approaches for solid block masonry exist using two-dimensional models [4,5], or three-dimensional models [6,7].

Another set of publications is dedicated to hollow block masonry made out of concrete or fired clay. Again there are two-dimensional approaches [8] as well as three-dimensional approaches [9–12]. Especially in Central Europe, vertically perforated clay block masonry is used extensively, particularly in residential buildings. Such vertically perforated clay blocks have a void pattern formed by orthogonally oriented webs. These webs can be divided into two groups according to their orientation: *longitudinal webs* are oriented parallel to the wall surface; *transversal webs* are oriented orthogonal to the wall surface (see Fig. 1). To our knowledge, the approach introduced by Nguyen and Meftah [11] is currently the only published model which covers the requirements for this kind of masonry. In a three-dimensional coupled temperature–displacement finite-element analysis, they simulated four full-scale fire experiments with good agreement of the temperatures and displacements. Nguyen and Meftah thoroughly investigated the mechanisms driving the failure of vertically perforated clay block masonry in fire experiments and identified the progressive spalling of the longitudinal webs as the most relevant [13,14]. This mechanism

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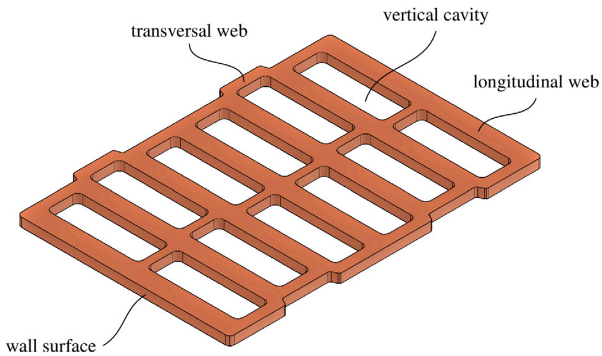


Fig. 1. Parts of a vertically perforated clay block.

refers to the detachment of parts of the masonry wall, which leads to a decreasing wall thickness. With vertical loads present, spalling is accompanied by an increasing load eccentricity, which introduces significant bending moments. In their studies, they suggested three mechanisms driving spalling:

- detachment of longitudinal webs due to tensile cracks at the intersection between longitudinal webs and transversal webs,
- buckling of longitudinal webs due to compressive stresses, and
- crushing of longitudinal webs due to a network of compressive cracks.

Therefore, they proposed a spalling criterion based on a maximum stress condition and plate buckling equations and introduced it to their finite-element framework.

Especially the first mechanism, i. e. the interaction between transversal and longitudinal webs, is essential for the mechanical behavior in fire situations for the following reason: In the case of a fire, the longitudinal web closest to the fire gets heated up. This increase in temperature causes a significant increase in strains. Since the temperature in the second longitudinal web increases delayed to the first one, these two webs show a difference in temperature-induced strains. These strains would not induce any stresses in an internally static determinate system, but this is not the case with a typical block design. Some design features, such as the type of head joint, the masonry bond, and the transversal webs, restrict the thermally-induced strains, inflicting stresses in different regions of the block.

In the case of fire, the mentioned strain difference of the longitudinal webs induces significant stresses in the transversal webs, which will start failing at some point. As soon as a certain amount of transversal webs has failed, the outermost longitudinal web spalls due to the vertical loading [13]. As more and more longitudinal webs fail, the eccentricity of the vertical load increases until the whole wall collapses eventually.

While the model of Nguyen and Meftah [11] can simulate a fire experiment with good results, we wanted to take a more detailed look at the stresses and damage appearing in the blocks, requiring a finer mesh. Especially when comparing different block designs or optimizing a product, these details can be essential. Since Nguyen and Meftah modeled an entire wall specimen, a finer mesh would lead to exploding computation times. Therefore, we used a *unit-cell approach* to model only a small representative part of the specimen with *periodic boundary conditions*. The saved computational expense could be reinvested in more detailed meshing and sophisticated non-linear material models. Unfortunately, the overall deformation of the wall cannot be obtained using this approach. Since the model published by Nguyen and Meftah already does a great job in calculating the deformations, the main aim of our study should be in another field of interest: the estimation of the endurance of load-bearing vertically perforated clay brick masonry in fire situations with a focus on the stress and strain fields, as well as damage in the cross-section.

Thus, the outline of the work can be summarized as follows: First, we created a two-dimensional transient thermal finite-element model to simulate the non-linear heat transfer in a wall made of vertically perforated clay block masonry. Secondly, we defined a novel *energy-based spalling criterion* to predict the failure of the wall in terms of spalling of longitudinal webs. Then we applied the nodal temperatures gained from the thermal model to a two-dimensional mechanical finite-element model. Using *Concrete Damaged Plasticity (CDP)*, a *unit-cell approach* with *periodic boundary conditions*, and the *energy-based spalling criterion*, this model was able to reproduce the behavior of a masonry wall in a standard fire test, without using any empirical fitting parameters. We also used a three-dimensional mechanical finite-element model to observe the stresses introduced by the vertical loading applied in the experiment. Given the model's high level of detail, the proposed approach provides unique insights into the behavior of fired clay block masonry in fire situations.

Section 2 contains a short description of the used experiments. An overview of the applied modeling strategies, the numerical model, and the used material properties is provided in Section 3. Afterward, the results are explained and discussed in Section 4, followed by conclusions to these results in Section 5.

2. Fire experiments on vertically perforated clay block masonry

2.1. Test setup

The tested masonry wall specimen was seven blocks high and six blocks wide, resulting in an approximately 3.00 m high and 3.00 m wide specimen (see Fig. 2(a)). This specimen was placed in a concrete frame with gaps on the left and right sides; these gaps were then filled with insulation. The blocks were connected by thin-bed mortar in the horizontal joints. The two adjacent blocks were in contact in the vertical joints, but no mortar was used. The blocks themselves had a thickness of 200 mm with the void pattern shown in Fig. 2(b).

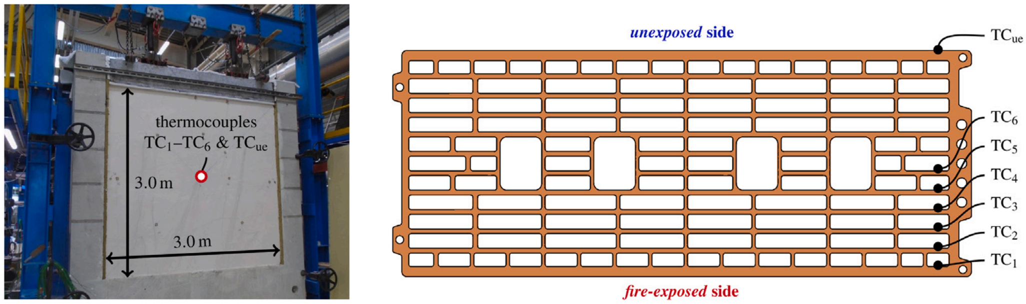
While the *unexposed* side was plastered, the *fire-exposed* side was covered with 120 mm expanded polystyrene (EPS) and a 13 mm gypsum plasterboard.

Thermal and mechanical loads were applied, as described next. The wall was subjected to vertical compressive loads amounting to 130 kN/m (load-controlled application using hydraulic presses), which corresponds to approximately 10 % of the compressive masonry strength. This load was applied to a loading bar resting on top of the specimen and kept constant throughout the experiment. An additional construction guided the loading bar to minimize out-of-plane deformations (see Fig. 2(a)). The plasterboard side of the wall faced a furnace in which the temperature was increased according to the temperature curve in EN 13501-2 [2] (similar to the curve in ISO 834-1 [15]) reading as

$$T_{\text{furnace}} = 345 \cdot \log_{10}(8t + 1) + 20, \tag{1}$$

with the time t starting at the beginning of the experiment (see also Fig. 3).

The temperature was monitored at multiple locations, i. e. the furnace, the cavities, and the unexposed face of the wall. The thermocouples relevant to the simulations were positioned along the thickness direction inside six different adjacent cavities of the block and on the unexposed side, all located in the center of the wall (see points labeled TC_1 to TC_6 as well as TC_{ue} in Fig. 2).



(a) (b)

Fig. 2. Masonry specimen (a) subjected to vertical compressive load. The Block geometry (b) is composed of many slender webs. The seven thermocouples (TC) relevant for the simulations were sitting in six adjacent cavities and on the *unexposed* side, all located in the center of the wall.

2.2. Test results: temperature evolutions and spalling times

The moment when the plasterboard fell off and the EPS burned down, i. e. the moment when the blocks were exposed to the thermal load, was considered the actual start of the test for the simulations, indicated by $t = 0$. The times referred to are always shown in relation to the total time, i. e. $\tau = \frac{t}{t_{\max}}$. While the temperature did not increase significantly on the unexposed side of the wall (see thermocouple $\mathrm{TC_{ue}}$ in Fig. 3), the temperature in the cavities increased in sequential fashion (see Fig. 3) due to sequential spalling of the longitudinal webs. After spalling of the first web, a rapid increase of the temperature T_{TC1} in the outermost cavity could be observed, quickly followed by a similarly rapid increase of T_{TC2}. The temperature T_{TC3} increased significantly only after roughly $\tau = 0.116$, the increase for temperature T_{TC4} became significant around $\tau = 0.290$. The temperature in the fifth outermost cavity T_{TC5} increased significantly only after roughly $\tau = 0.362$, but the increase was already less sharp. The temperature T_{TC6} remained below 120 °C, even after $\tau = 1.0$, when the wall finally collapsed.

Deformations towards the furnace could be observed, with a maximum value at approximately $\frac{2}{3}$ of the specimen height. These deformations can be traced back to the material expansion at the fire-exposed side in combination with the boundary conditions and are already well-documented in literature (e. g. Nguyen and Meftah [13], Prakash et al. [5]). According to these publications, the bottom edge can be considered as clamped, which leads to bending moments in the wall. From the center of the wall downward, these bending moments yield vertical compressive stresses on the fire-exposed side, causing the most severe degradation of the blocks at these locations. Therefore, we used the thermocouples located in the center for evaluating the results and validating the numerical model.

As discussed next, we estimated the times when the longitudinal webs spalled in the experiment from the cavity temperature gradients (Fig. 3). This spalling time t_i of the longitudinal web i was estimated by the sharp increase of the temperature derivative in the next cavity measured with thermocouple TC_{i+1}. Since the mass of the heated longitudinal web delays the temperature increase in the cavity behind, spalling was considered to occur 15 s before the temperature change reached 3 °C/s (the dashed line indicates this threshold in Fig. 3). For t_5, this criterion could not be applied since the temperature change of thermocouple $\mathrm{TC_6}$ stayed below this threshold. Given the increasing gradient of T_{TC6} at the end of the experiment, t_5 was chosen 10 s before t_6, where the wall collapsed. The estimated spalling times are summarized in Table 1, and they were used to divide the experiment into seven Phases (0 to 6), which are dealt with individually in the following modeling section. Notably, these times were solely used for validating the thermal modeling strategies; in the simulations, an energy-based criterion was used to obtain spalling times from a mechanical finite-element model.

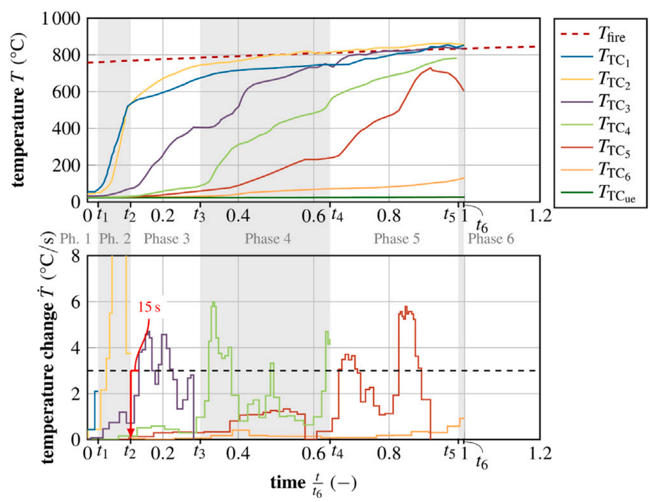


Fig. 3. Evolution of cavity temperatures T_i (top) and evolution of temperature change $\dot{T}_i$ (bottom). The spalling times t_i for each phase were estimated 15 s before the temperature change reached 3 °C/s, as shown for t_2. The respective curves are cut off as soon as a thermocouple is considered exposed to the furnace temperature. The furnace temperature T_{fire} was derived according to EN 13501-2 [2] as well as ISO 834-1 [15].

3. Numerical modeling

3.1. Sequential spalling and associated modeling phases

The sequential spalling of the longitudinal webs leads to a geometry change and a corresponding change in the boundary conditions. Hence, the numerical modeling of the fire experiment was divided into seven phases introduced in Table 1. Note that the spalling times obtained from the experiment were not used for the numerical model (except for validating the 2D thermal model in Section 4.1). Instead, an energy-based spalling criterion was used to obtain a spalling time for each phase of the numerical model. Phase 0 was considered irrelevant for the model since the cavity temperatures remained very low due to the thermal insulation provided by the intact EPS cover (see Section 2). As for Phase 1, the initial block geometry was used. In Phase 2, the first longitudinal web and all transversal webs in the first cavity were removed (Fig. 4), leading to the exposure of the second longitudinal web to the fire. This strategy was extended to all following phases until Phase 6, where the first five longitudinal webs and the corresponding parts of the transversal webs were removed.

Table 1
Subdivision of the experiment into seven Phases according to the experimentally observed spalling times.

Phase 0		Start of the experiment, wall loaded with 130 kN/m
	$\tau_0 = \frac{t_0}{t_6} = 0.000$	Gypsum plasterboard came down, EPS burned down, block surface exposed to fire
Phase 1		
	$\tau_1 = \frac{t_1}{t_6} = 0.028$	First longitudinal web spalled, second web exposed to fire
Phase 2		
	$\tau_2 = \frac{t_2}{t_6} = 0.115$	Second longitudinal web spalled, third web exposed to fire
Phase 3		
	$\tau_3 = \frac{t_3}{t_6} = 0.300$	Third longitudinal web spalled, fourth web exposed to fire
Phase 4		
	$\tau_4 = \frac{t_4}{t_6} = 0.645$	Fourth longitudinal web spalled. Fifth web exposed to fire
Phase 5		
	$\tau_5 = \frac{t_5}{t_6} = 0.986$	Fifth longitudinal web spalled, sixth web exposed to fire
Phase 6	$\tau_6 = \frac{t_6}{t_6} = 1.000$	Wall collapsed

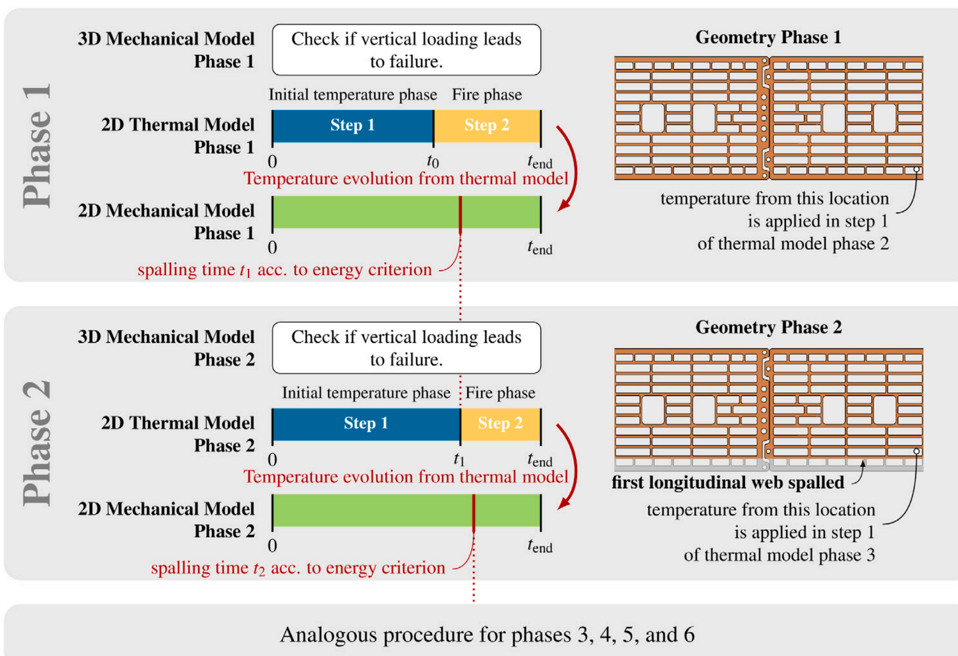


Fig. 4. Overview of the sequential procedure for modeling the six-phase fire experiment (Phases 1 to 6, experimental Phase 0 is omitted): Each phase consists of an elastic 3D mechanical model, a two-step 2D thermal, and a 2D mechanical model. The nodal temperatures obtained in the thermal model of Phase i were imposed on the mesh of the corresponding mechanical model of Phase i. The spalling time predicted in the mechanical model of Phase i sets the duration of the steps in the thermal model of the subsequent Phase $i + 1$.

3.2. Decoupling of fire loading from vertical loading and corresponding unit-cell models

Previous fire experiments of masonry without vertical loading, see e. g. Nguyen and Meftah [13], indicate no spalling of longitudinal webs and no wall failure until the end of the experiment but significant cracking in transversal webs. In contrast, tests with vertical loading, like the test presented in Section 2, show prevalent spalling of longitudinal webs after the cracking of the adjacent transversal webs. Notably, the sequential spalling of the webs introduces an eccentricity of the vertical loading (see Fig. 6(b)), which leads to bending moments in the wall and thus to additional vertical stresses, which eventually lead to the collapse of the whole structure.

For both loaded and unloaded walls, failure of transversal webs due to temperature-induced stresses is crucial and was tackled in the first step, independently from the vertical loading. A two-dimensional finite-element model was used to solve two main problems: (i) to predict the transient temperature field in the brick wall for each phase of the fire test and (ii) to model the cracking of transversal webs due to temperature-induced stresses leading to spalling of the longitudinal webs. Notably, the modeling of the transient temperature field within a single phase is, in good approximation, *independent* of the mechanical solution since the deformations were generally small compared to the size of the structure and since stresses have no effect on the thermal properties of the materials. Therefore, the sub-problems mentioned above could be solved individually: A 2D thermal model was used to obtain the temperature field (Section 3.3), which was then applied to a 2D mechanical model (Section 3.4) to predict the stress field and eventually the spalling (see Fig. 4). The vertical loading and the load eccentricity were then considered separately within a 3D finite-element model (Section 3.5). Although three different FE models in each phase are necessary for this approach, the decoupling is still more efficient than a single 3D coupled approach with a similar mesh and material behavior.

A unit-cell approach was adopted to minimize the computational effort. In more detail, the 2D unit-cell exhibits a length of a single brick and contains two halves of the brick connected by a central joint (see Fig. 5). The corresponding 3D unit-cell is two blocks high and one block wide (see Fig. 6(a)) and is based on a recently developed FE model for failure of masonry Kiefer et al. [16].

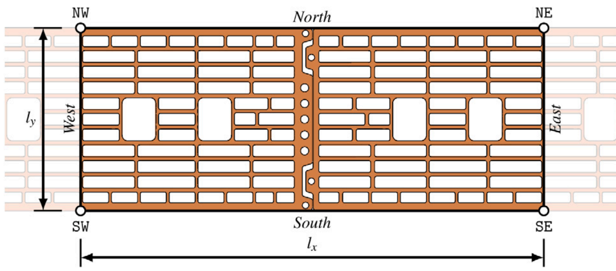


Fig. 5. Two-dimensional unit-cell model. The edges of the model were named *North*, *East*, *South*, and *West*; the Vertices were named after the edges intersecting in the particular vertex.

3.3. Two-dimensional transient thermal finite-element model

3.3.1. Strategy, mesh, and boundary conditions

A two-dimensional transient thermal finite-element simulation was performed in the FE software Abaqus to identify the temperature field over the entire duration of the experiment. Thereby, both the solid clay parts and the air-filled cavities were considered in the model. For the latter, a temperature-dependent effective thermal conductivity was introduced to model convective and radiant heat transfer. Conductive heat transfer can be described by the heat conduction equation, which reads as

$$(\rho \cdot c_{\mathrm{p}}) \cdot \dot{T} - \nabla(\lambda \cdot \nabla T) = 0, \tag{2}$$

with the mass density ρ, the heat capacity c_{p}, the temperature T, and the thermal conductivity tensor λ [17]. In the case of isotropic conductivity, the thermal conductivity tensor is formed by multiplying the thermal conductivity λ with the identity matrix $\mathbf{I}$.

Each thermal model consists of two loading steps (see Fig. 4). In both steps, adiabatic boundary conditions were applied on the *East* and *West* side of the model. On the "unexposed" side of the model (*North*), the temperature was fixed at 21 °C in both steps. To recreate the temperature and heat flux field at the end of the previous Phase $i-1$ in Phase i, the temperature $T_{\mathrm{TC}i-1}$ (see Fig. 2) was applied on the *fire-exposed* side of the model (*South*) in the first step. In the second step, the furnace temperature after Eq. (1) [2,15] was applied directly on the fire-exposed side of the model. The boundary conditions can be found in Fig. 7. The finite-element models consist of 14676 to 29183 three-node and four-node linear diffusive heat transfer elements (DC2D3 and DC2D4, respectively), see Fig. 7.

3.3.2. Thermal properties of fired clay

The mass density of the fired clay in the experiments was $\rho_{\mathrm{c}} = 1575\,\mathrm{kg/m^3}$ and was considered to be temperature-independent, see also Table 2. In contrast to the mass density, the thermal conductivity and specific heat were considered temperature-dependent [14,18]. At room temperature, they amount to $\lambda_{\mathrm{c}} = 0.42\,\mathrm{W/(m\cdot K)}$ (calculated after EN 1745 [19]) and $c_{\mathrm{p,c}} = 876\,\mathrm{J/(kg\cdot K)}$ [11]. From steady-state heat flow experiments on hollow clay blocks, Nguyen et al. [14] estimated a relatively constant thermal conductivity of approximately 300 °C followed by a linear decrease to 39 % of the initial value at 800 °C. However, calculating the thermal conductivity from measurements on the mass density, the specific heat, and the thermal diffusivity, AIT [18] obtained a nearly linear decrease of the thermal conductivity up to 400 °C. A linear decrease from 21 to 800 °C was used for the numerical simulations (see Fig. 8(a)), considering these observations. Notably, running the model with both evolutions suggested in the literature led to nearly identical temperature results.

The specific heat of fired clay increases slightly with increasing temperature [18], whereby a significant peak is considered around 100 °C (see Fig. 8(b)). Using this relation, we followed the suggestions in the EN 1996-1-2 [20] standard, and the modeling strategies adopted by Nguyen et al. [14] for clay blocks as well as Prakash et al. [5] for concrete blocks (see Fig. 8(b)). The corresponding increase, $\Delta c_{p,c}^{\mathrm{peak}} = 2825\,\mathrm{J/(kg\cdot K)}$, results from the evaporation of water bound in the pores of the brick and is obtained from the brick's water content $\omega_{\mathrm{c}} = 1.25\,\%$ as [14]:

$$\Delta c_{p,c}^{\mathrm{peak}} = \frac{2 \cdot \omega_{\mathrm{c}} \cdot H_{\mathrm{vap}}}{\Delta T_{\mathrm{peak}}} \tag{3}$$

with the latent heat of vaporization of water, $H_{\mathrm{vap}} = 2260\,\mathrm{kJ/kg}$, and with $\Delta T_{\mathrm{peak}} = 20\,\mathrm{K}$ as the temperature interval for the peak. This way, the heat capacity at 100 °C is approximately four times higher than the initial value (see Fig. 8(b)).

3.3.3. Heat transfer in the cavities — conduction, convection, and radiation

Heat transfer is driven by three mechanisms: *conduction*, *convection*, and *radiation*. While conduction dominates in solids, convection and radiation are decisive in gases.

Neither convective nor radiant heat transfer in the air cavities was modeled directly. Instead, we considered convection and radiation through an effective conductivity, $\lambda_{\mathrm{eff,rad}}$ and $\lambda_{\mathrm{eff,conv}}$ respectively, as this is often done when calculating the thermal resistance of ventilated air-spaces behind façades [25]. Adding these quantities to the thermal conductivity λ_{a} leads to an overall effective conductivity λ_{eff},

$$\lambda_{\mathrm{eff}} = \lambda_{\mathrm{a}} + \lambda_{\mathrm{eff,rad}} + \lambda_{\mathrm{eff,conv}}, \tag{4}$$

which was used in Abaqus within a purely conductive transient heat transfer simulation based on Eq. (2). This strategy comes with an inaccuracy compared to the "real" problem. However, we also wanted the same model to be fit for simulating clay block masonry with insulation-filled cavities. In a comparative analysis, the temperature distributions only varied by a maximum of 20 °C

The effective radiant heat transfer is dealt with first. Since the heat flow in the given problem is nearly one-dimensional (i. e. in y-direction), only the surfaces parallel to the wall were considered when calculating the effective conductivity. Radiation of transversal webs was neglected (see Fig. 9). The radiant heat flow q_{rad} between two surfaces, which dimensions are significantly larger than their distance, follows from the Stefan–Boltzmann law as ([26, p. 25])

$$q_{\mathrm{rad}} = \frac{\sigma}{\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1} \cdot \left(T_1{}^4 - T_2{}^4\right). \tag{5}$$

Hereby, σ is the Stefan–Boltzmann constant ($5.67 \times 10^{-8}\,\mathrm{W/(m^2K^4)}$), ε_i is the emissivity of surface i, and T_i is the surface temperature of surface i. The surfaces are denoted 1 and 2, as shown in Fig. 9. The emissivity was set to 0.9 for both surfaces [11,22]. The corresponding one-dimensional conductive heat flow within the air void follows from Fourier's law of heat conduction in the form

$$q_{\mathrm{eff,rad}} = \frac{\lambda_{\mathrm{eff,rad}}}{d} \cdot \left(T_1 - T_2\right), \tag{6}$$

where $\lambda_{\mathrm{eff,rad}}$ is the effective radiant thermal conductivity and d is the layer's thickness. Equating Eqs. (5) and (6) allows for the determination of the sought effective conductivity of the voids, reading as

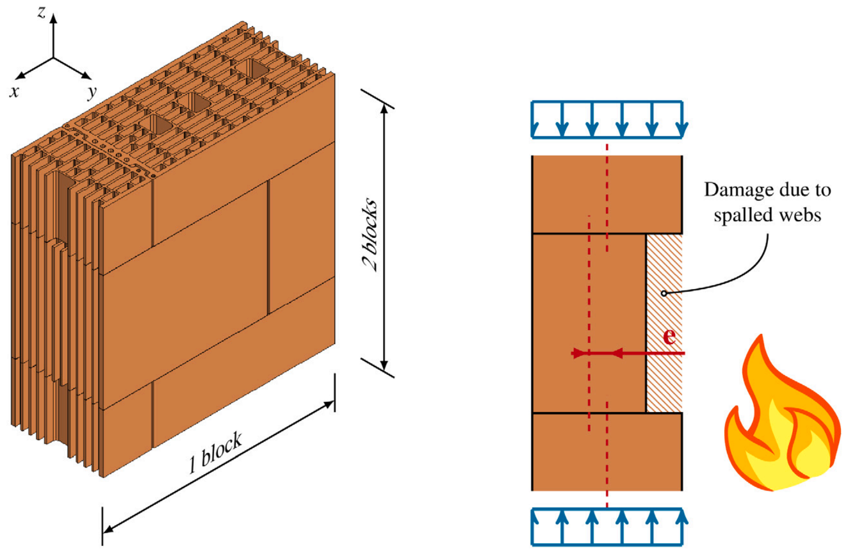
$$\lambda_{\mathrm{eff,rad}} = \frac{\sigma \cdot d}{\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1} \cdot \frac{T_1{}^4 - T_2{}^4}{T_1 - T_2}. \tag{7}$$

Convective heat flow depends on the thermal conductivity of the gas, λ_{a}, and the so-called Nusselt number Nu, which describes the ratio of the amount of heat transported in a flowing medium, compared to a static one:

$$\lambda_{\mathrm{eff,conv}} = Nu \cdot \lambda_{\mathrm{a}}. \tag{8}$$

For calculating the Nusselt number, the cavities were considered as vertical shafts, which are heated from one side. The VDI Heat Atlas pp. 681–682 [21], provides the following equation for this case:

$$Nu = \left(\left(0.08333 \cdot Ra \cdot \frac{d}{H}\right)^{-\frac{3}{2}} + \left(0.61 \cdot \left(Ra \cdot \frac{d}{H}\right)^{\frac{1}{4}}\right)^{-\frac{3}{2}} \right)^{-\frac{2}{3}}. \tag{9}$$



(a) (b)

Fig. 6. Three-dimensional mechanical finite-element model (a) and load eccentricity (b), which is introduced by the spalling longitudinal webs.

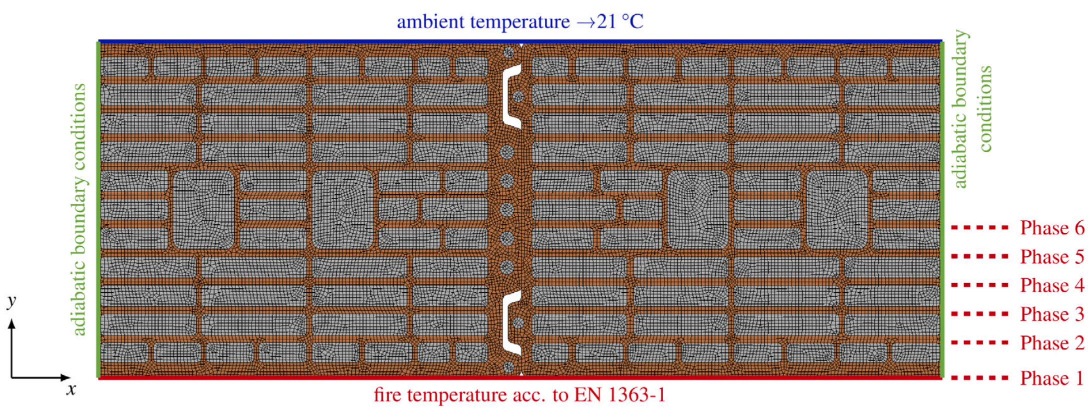
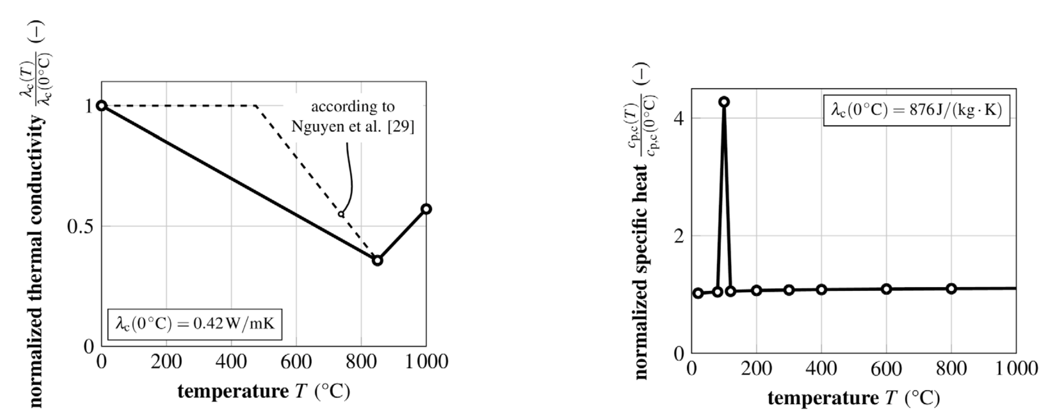


Fig. 7. The two-dimensional thermal finite-element model for the first experimental phase (all webs intact) with the imposed boundary conditions. The dashed lines indicate where the geometry was cut for later phases and, therefore, where the fire temperature was applied.



(a) (b)

Fig. 8. Temperature-dependent thermal properties of fired clay used in the finite-element model: (a) thermal conductivity from a combination of results from [18], and [14], (b) specific heat after [14].

Table 2
Thermo-mechanical properties of air, fired clay, and mortar at $T_{ref} = 21\,°C$.

Air		temp. -dep.	Reference	Fired clay		temp. -dep.	Reference
Mass density ρ_a	$1.2\,kg/m^3$	✓	Fig. 10(e), [21]	Mass density ρ_c	$1575\,kg/m^3$	×	–
Specific heat $c_{p,a}$	$1007\,J/(kg·K)$	✓	Fig. 10(d), [21]	Specific heat $c_{p,c}$	$876\,J/(kg·K)$	✓	Fig. 8(b), [14]
Thermal cond. λ_a	$0.024\,W/(m·K)$	✓	Fig. 10(c), [21]	Thermal cond. λ_c	$0.42\,W/(m·K)$	✓	Fig. 8(a), [14,18]
Kin. viscosity ν_a	$1.59 \times 10^{-5}\,m^2/s$	✓	Fig. 10(a), [21]	Emissivity ε_c	0.9	×	[11,22]
Dyn. viscosity η_a	$1.82 \times 10^{-5}\,kg/(m·s)$	✓	Fig. 10(b), [21]	CTE $\alpha_{T,c}$	$8 \times 10^{-6}\,T^{-1}$	✓	Fig. 16(b), [23]
				Young's mod. E_c	10 220 MPa	✓	Fig. 16(a), [14,16]
Mortar				Poisson's ratio ν_c	0.2	×	[16]
Young's mod. E_m	5000 MPa	×	[16]	Tensile strength $\sigma_{f,t,c}$	7.5 MPa	×	Fig. 14(a), [16,24]
Poisson's ratio ν_m	0.1	×	[16]	Comp. strength $\sigma_{f,c,c}$	27.6 MPa	×	Fig. 14(b), [16]

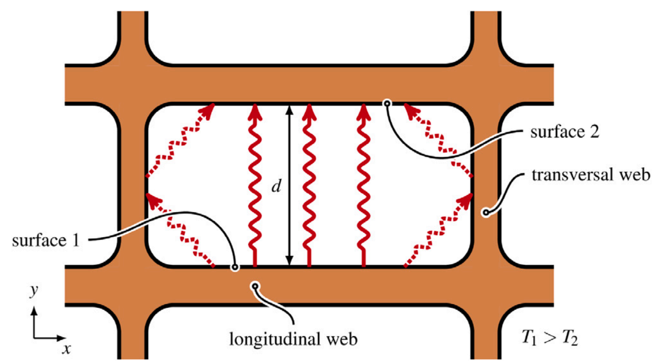


Fig. 9. Horizontal section of vertically perforated brick: Governing paths of radiant heat transfer. The thermal interaction of the transversal webs due to radiation (dashed paths) was neglected.

Hereby, d and H are the cavity's thickness and height, respectively. The Rayleigh number Ra is defined by

$$Ra = \frac{g \cdot (T_1 - T_m) \cdot d^3}{T_m \cdot \nu_a{}^2} \cdot Pr \qquad \text{with the Prandtl number } Pr = \frac{\eta_a \cdot c_{p,a}}{\lambda_a}, \tag{10}$$

with the gravitational constant g ($9.81\,m/s^2$), the mean temperature of the air in the cavity, T_m, and the distance between the cavity surfaces, d. ν_a, η_a, $c_{p,a}$, and λ_a are the kinematic viscosity, the dynamic viscosity, the heat capacity, and the thermal conductivity of air, respectively. The aforementioned properties were obtained for temperatures from 0 °C to 1000 °C according to the VDI Heat Atlas pp. 302–393 [21] (see Fig. 10 and Table 2).

For calculating the effective thermal conductivity according to Eqs. (4) and (7)–(10), the cavities were sorted into four groups, considering their thickness d (6 mm, 8 mm, 14 mm, and 38 mm). The height of the cavities amounts to $H = 3.0\,m$, assuming that they form a connected shaft from the bottom to the top of the wall. Fig. 11 shows the calculated effective thermal conductivity for the cavity groups. Since the heat transfer at higher temperatures is governed by radiation and the distance of the surfaces has nearly no impact on the radiant heat transfer, the effective conductivity increases significantly with increasing thickness (e. g. Group 2, which contains the cavities in the middle with the largest thickness). For the smaller cavities, which are outnumbering the larger ones, the effective thermal conductivity λ_{eff} exceeds the value of fired clay around 200 °C–300 °C.

The temperature-dependent mass density of air, ρ_a was also taken from the VDI Heat Atlas pp. 302–393 [21]. A significant decrease in mass density with increasing temperature can be observed (see Fig. 10(e)).

Since the effective conductivity depends on the surface temperatures in the cavities, and these surface temperatures, in turn, depend on the effective conductivity, an iteration process seems to be necessary. However, evaluating Eqs. (8)–(10) for different temperature differences and different cavity sizes showed that the influence of the temperature difference between the surfaces is insignificant. In contrast, the distance between the surfaces has a significant impact (see Fig. 12). Hence, the heat transfer was calculated without iteration using a fixed temperature difference of 50 °C.

3.4. Two-dimensional mechanical finite-element model

3.4.1. Strategy, mesh, and boundary conditions

By analogy to the thermal model discussed before, geometrically identical two-dimensional mechanical finite-element models (see Fig. 13) were created for each of the experimental phases (see Fig. 4). These models were used to evaluate the stress fields resulting from the temperature fields calculated with the thermal model and to predict the sequential spalling of the longitudinal webs. The models consist of 3885 to 6336 three-node and four-node linear plane stress elements (CPS3 and CPS4, respectively). A plane stress model with vanishing vertical stresses $\sigma_{zz} = 0$ was considered since the dead weight was neglected, and the vertical load was only considered in the three-dimensional model. In contrast to the thermal models, elements in the cavities were not needed, and the mesh was only refined in the first two longitudinal webs to shorten computation times.

(*Periodic boundary conditions (PBC)* were considered to couple the displacements on the West and East boundary of the unit-cell (see Fig. 5). In contrast, *North* and *South* boundaries may deform freely. This way, a periodic (in longitudinal x direction) structure is modeled, which is comparable to the tested wall, given that the wall length is more than one order of magnitude larger than the wall thickness.

The linear equations, which couple the displacements of the *East* and *West*, were derived as described next. The displacements u_x and u_y of each point on the *East* boundary can then be defined by the displacements of the *vertex nodes* on this boundary (NE, SE, NW or SW, see Fig. 5), the length of the model in y-direction, l_y, and the location of the point:

$$\begin{pmatrix} u_x^E(y) \\ u_y^E(y) \end{pmatrix} = \begin{pmatrix} u_x^{SE} \\ u_y^{SE} \end{pmatrix} + \frac{y}{l_y} \cdot \begin{pmatrix} u_x^{NE} - u_x^{SE} \\ u_y^{NE} - u_y^{SE} \end{pmatrix}. \tag{11}$$

The same relation applies on the *West* boundary:

$$\begin{pmatrix} u_x^W(y) \\ u_y^W(y) \end{pmatrix} = \begin{pmatrix} u_x^{SW} \\ u_y^{SW} \end{pmatrix} + \frac{y}{l_y} \cdot \begin{pmatrix} u_x^{NW} - u_x^{SW} \\ u_y^{NW} - u_y^{SW} \end{pmatrix}. \tag{12}$$

Forming the difference of Eqs. (11) and (12) results in the coupling of the edges *East* and *West*:

$$\begin{pmatrix} \Delta u_x \\ \Delta u_y \end{pmatrix} = \begin{pmatrix} u_x^E(y) - u_x^W(y) \\ u_y^E(y) - u_y^W(y) \end{pmatrix} = \frac{l_y - y}{l_y} \cdot \begin{pmatrix} u_x^{SE} - u_x^{SW} \\ u_y^{SE} - u_y^{SW} \end{pmatrix} + \frac{y}{l_y} \cdot \begin{pmatrix} u_x^{NE} - u_x^{SW} \\ u_y^{NE} - u_y^{SW} \end{pmatrix}. \tag{13}$$

The linear Eq. (13) was applied to each pair of corresponding nodes to ensure the geometrical compatibility of the unit-cell. For applying the PBCs to the model, we developed the free-to-use module *AbaqusUnitCell2D* [27] for the scientific programming language *Julia* (see Bezanson et al. [28]), which works for Abaqus input files. Notably, a 3D counterpart of these equations can be found in Suda et al. [29].

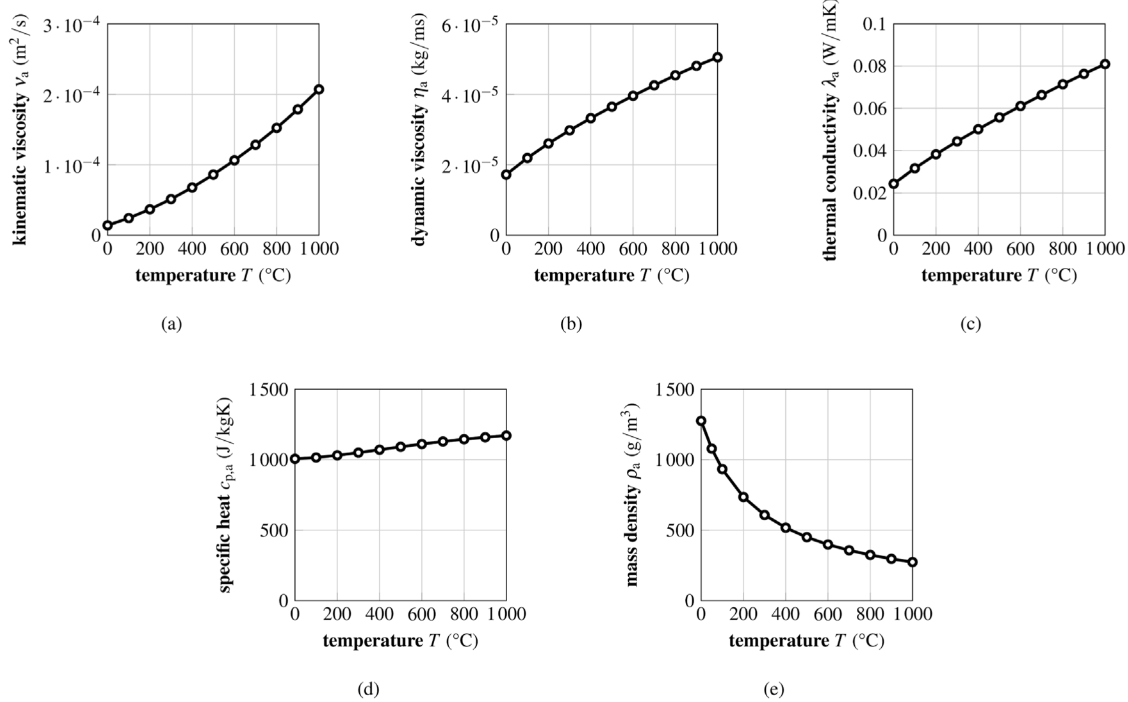


Fig. 10. Temperature-dependent thermal properties of air used in the finite-element model: (a) kinematic viscosity ν_a, (b) dynamic viscosity η_a, (c) thermal conductivity λ_a, (d) specific heat $c_{p,a}$, and (e) mass density ρ_a [21].

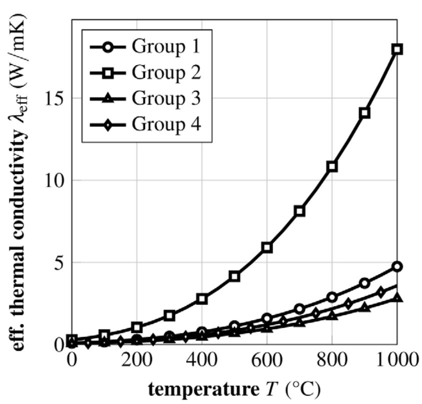


Fig. 11. Effective thermal conductivity of the cavities, λ_{eff}, subdivided into four groups, depending on the cavity thickness: 14 mm (Group 1), 38 mm (Group 2), 6 mm (Group 3), and 8 mm (Group 4). The values were derived using Eqs. (4) to (10).

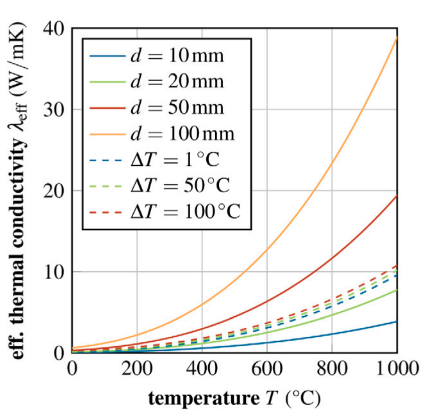


Fig. 12. Case study for estimating the impact of temperature difference $\Delta T = T_1 - T_2$ (with $d = 25$ mm) and distance between the cavity surfaces d (with $\Delta T = 10$ °C) on the effective thermal conductivity λ_{eff} of the cavities.

Additionally, springs were introduced at the four vertices (labeled NW, NE, SW, SE in Fig. 13) to consider the stiffening effect, which originates in the offset of the upper and lower block layer. This offset leads to a reduction of the horizontal in-plane deformations (x-direction). The spring stiffness (300 N/m) was chosen in such a way that the estimated spalling time for Modeling Phase 1 corresponded with the experimental observations. For the subsequent phases, this value was not changed. Apart from the kinematic boundary conditions, the temperature field of the associated thermal model was applied to the whole mechanical model. The boundary conditions can be found in Fig. 13.

3.4.2. *Constitutive behavior adopted for numerical simulations*

The stresses are coupled to the temperatures in the constitutive law (see Mang and Hofstetter [30])

$$\sigma_{ij} = \frac{E}{1+\nu} \cdot \left(\varepsilon_{ij} + \frac{\nu}{1-2\nu} \cdot \varepsilon_{kk} \cdot \delta_{ij} \right) - \frac{E}{1-2\nu} \cdot \alpha_T \cdot \left(T - T_0\right) \cdot \delta_{ij} \tag{14}$$

with the nine independent components of the stress tensor, σ_{ij}, the material's Young's modulus, Poisson's ratio, and thermal expansion coefficient, E, ν, and α_t respectively, the nine independent components of the strain tensor, ε_{ij}, the Kronecker delta δ_{ij}, and the current temperature T as well as the initial temperature T_0. Eq. (14) only describes the linear elastic constitutive law; material non-linearities were considered by implementing *Concrete Damaged Plasticity (CDP)*.

CDP is a plasticity-based smeared-damage model developed for concrete, suitable for use with quasi-brittle materials. Iuorio and Dauda [31] as well as Silva et al. [32] already successfully applied CDP for fired clay. The model considers two main failure mechanisms: cracking under tensile and crushing under compressive stresses. Two hardening variables control the evolution of the yield surface for tension and compression separately, $\tilde{\varepsilon}_t^{pl}$ and $\tilde{\varepsilon}_c^{pl}$ respectively. Therefore, it is possible to define different softening behavior for the tensile and compressive regimes. The yield function implemented in Abaqus (see Simulia [33]) is a function proposed by Lubliner et al. [34] with adaptations made by Lee and Fenves [35]:

$$F = \frac{1}{1-\alpha} \left(\bar{q} - 3\alpha \cdot \bar{p} + \beta \left(\tilde{\varepsilon}^{pl} \right) \cdot \langle \hat{\bar{\sigma}}_{max} \rangle - \gamma \cdot \langle -\hat{\bar{\sigma}}_{max} \rangle \right) - \bar{\sigma}_c(\tilde{\varepsilon}_c^{pl}) = 0, \tag{15}$$

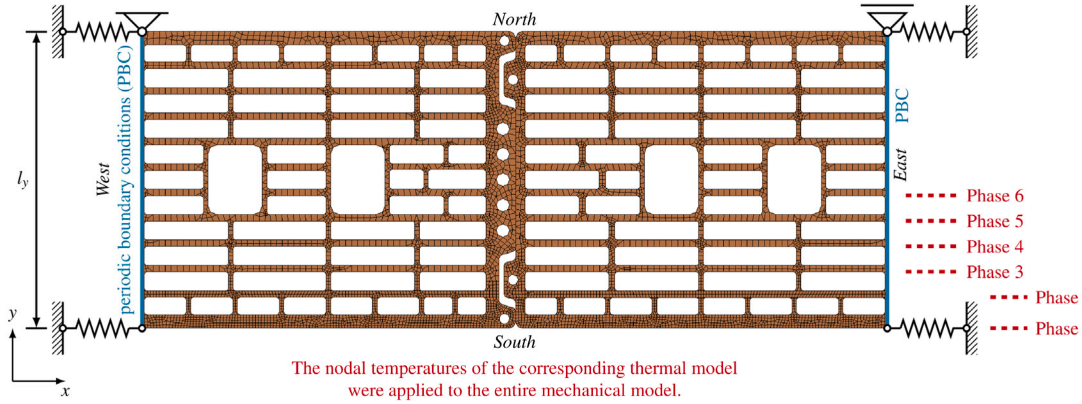


Fig. 13. The two-dimensional mechanical finite-element model for the first experimental phase (all webs intact) with the imposed boundary conditions. The dashed lines indicate where the geometry was cut for later phases and, therefore, where the fire temperature was applied.

with the hydrostatic pressure stress $\bar{p}$, the Mises equivalent stress $\bar{q}$, and the maximum principal effective stress $\hat{\bar{\sigma}}_{\max}$. These stress values are derived from the effective stress tensor $\bar{\boldsymbol{\sigma}}$:

$$\bar{\boldsymbol{\sigma}} = \mathbf{D}_0^{\mathrm{el}} \cdot \left(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\mathrm{pl}}\right), \tag{16}$$

with the initial, undamaged elasticity tensor $\mathbf{D}_0^{\mathrm{el}}$, and the total as well as the plastic strain tensor, $\boldsymbol{\varepsilon}$ and $\boldsymbol{\varepsilon}^{\mathrm{pl}}$ respectively. This total strain tensor includes elastic and plastic strains but not temperature-induced strains. The three variables α, β, and γ are derived from the input parameters as follows:

$$\alpha = \frac{\sigma_{b0}/\sigma_{c0} - 1}{2 \cdot \sigma_{b0}/\sigma_{c0} - 1}, \quad \beta = \frac{\bar{\sigma}_{\mathrm{c}}\left(\tilde{\varepsilon}_{\mathrm{t}}^{\mathrm{pl}}\right)}{\bar{\sigma}_{\mathrm{t}}\left(\tilde{\varepsilon}_{\mathrm{t}}^{\mathrm{pl}}\right)} \cdot (1-\alpha) - (1+\alpha), \quad \text{and} \quad \gamma = \frac{3 \cdot \left(1 - K_{\mathrm{c}}\right)}{2 \cdot K_{\mathrm{c}} - 1}. \tag{17}$$

The parameter K_{c} controls the shape of the meridional plane of the yield surface and was set to $K_{\mathrm{c}} = 2/3$ to approximate the yield surface of Mohr–Coulomb's criterion [33]. The dilation angle ψ amounts to 5°. The remaining input parameters, namely the ratio of initial equibiaxial compressive yield stress to initial uniaxial compressive yield stress, σ_{b0}/σ_{c0}, the "eccentricity" of the flow potential, ϵ, and the viscosity parameter, μ, take the values 1.16, 0.1, and 0, respectively, according to Silva et al. [32].

The tensile softening for the CDP, $\bar{\sigma}_{\mathrm{t}}\left(\tilde{\varepsilon}_{\mathrm{t}}^{\mathrm{pl}}\right)$, was defined after Pluijm [24], who used a relation proposed by Hordijk and Reinhardt [36] for concrete (see Fig. 14(a)). The peak stress amounts to $\sigma_{\mathrm{f,t,c}} = 7.5\,\mathrm{MPa}$, which is the materials tensile strength discussed in Section 3.4.4. The plastic strain value at reaching the minimal stress was chosen in a way that the mode-I-fracture energy in the simulation would fit commonly accepted values for fired clay ($0.05\,\mathrm{J/mm^2}$, see e. g. [37–39]).

Since tensile stresses are the most crucial for the failure of vertically perforated fired clay blocks (see Kiefer et al. [16] and Suda et al. [29], as well as Nguyen and Meftah [11,13]), the main focus was the correct depiction of the tensile softening. Thus, for the compressive softening, $\bar{\sigma}_{\mathrm{c}}\left(\tilde{\varepsilon}_{\mathrm{c}}^{\mathrm{pl}}\right)$, consideration of a bilinear stress–strain relation with plateau stress $\sigma_{\mathrm{f,c,c}} = 27.6\,\mathrm{MPa}$ was sufficient (see Fig. 14(b) and Table 2). The compressive and tensile strengths used for CDP are discussed in more detail in Section 3.4.4.

3.4.3. *Energy-based spalling criterion and failure of the wall*

To complete the modeling of temperature-induced *sequential* failure of the longitudinal webs in the brick wall, as observed experimentally, a criterion for spalling of the longitudinal web closest to the furnace, as well as a criterion for overall wall failure, were required. For the spalling criterion, we recall that failure of the connection between the transversal and the longitudinal webs triggers spalling (see Nguyen and Meftah [13] as well as Nguyen and Meftah [11]). An elastic approach, where the stresses in these connections are observed, is inadequate since the first cracks in the plastic simulations led to a redistribution of stresses and not yet to wall collapse. Therefore, we needed a criterion that is able to find the critical point where a redistribution of the stresses yields spalling of the longitudinal web exposed to fire. Given the smeared damage approach in the framework of the adopted CDP (see Section 3.4.2), the model can predict damage localization accompanied by dissipative plastic deformations. We expect energy to dissipate progressively during the simulations following the progressive increase of temperature-induced stresses. After some time, we expect the damage to yield an abrupt increase in plastic deformations and thus dissipated energy, particularly in the connection of transversal and longitudinal webs, accompanied by a drop in elastically stored energy (strain energy). These energy changes indicate internal load redistribution, and we assume that spalling is occurring at this point in the experiment. After a spalling event, the model geometry is changed accordingly, and the next modeling phase is entered. The flowchart in Fig. 15 qualitatively shows the implementation of our energy-based spalling criterion.

The overall failure of the wall was found considering the elastic 3D mechanical simulations. In each phase, the tensile stresses in the transversal webs (see also Fig. 21) were compared to the material's tensile strength. When these stresses exceeded the tensile strength, we considered the last obtained spalling time t_i as the total time at wall failure t_{end} (see Fig. 15).

3.4.4. *Mechanical properties of fired clay*

While EN 1996-1-2 [20] suggests a linear decreasing Young's modulus with increasing temperature, experimental observations show a slightly different behavior (e. g. Nguyen et al. [14]). Nguyen et al. suggest a decrease to approximately 70 % of the initial value between 21 °C and 100 °C, while staying nearly constant at higher temperatures (see Fig. 16(a)). Between 800 °C and 1000 °C Young's modulus decreases again to 15 % of its initial value. This behavior was implemented in the finite-element simulations.

Fired clay in extruded clay blocks shows a significant orthotropic behavior originating in the extrusion process [40]. Extensive information on the orthotropic microstructure of fired clay and the corresponding mechanical modeling can be found in the research of Kariem et al. [41,42] and Buchner et al. [43,44]. The plate-like shape of the webs causes the clay minerals to be oriented parallel to the web's surface. Therefore, Young's modulus parallel to the web's surface is approximately 100 %–200 % larger than orthogonal to it, depending on the composition of the clay material [45]. Unfortunately, using CDP as plasticity law does not allow anisotropic material stiffness. Since the web length is by 5 to 18 times larger than their thickness, the occurring in-plane stresses are significantly larger than the out-of-plane stresses. Hence, the larger in-plane value of Young's modulus was used, which amounts to $E_{\mathrm{c}} = 13\,500\,\mathrm{MPa}$ at $T_{\mathrm{ref}} = 21\,°\mathrm{C}$.

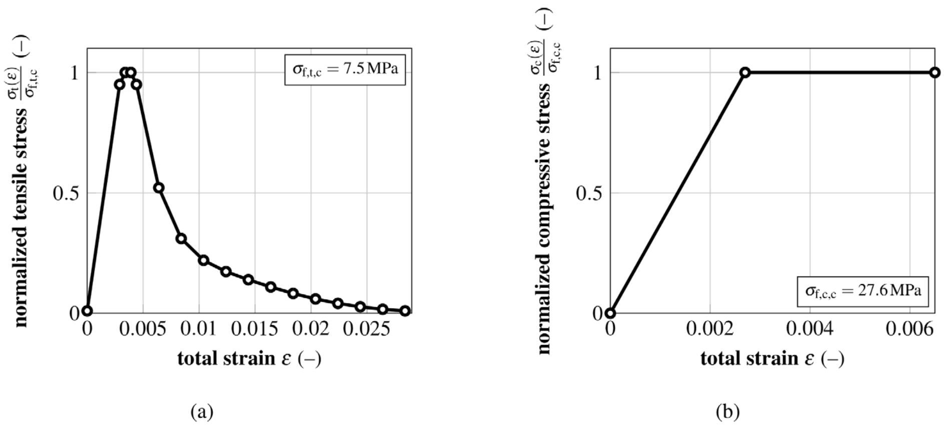


Fig. 14. Adopted softening behavior for fired clay in the tensile regime (a) [24,36] and compressive regime (b).

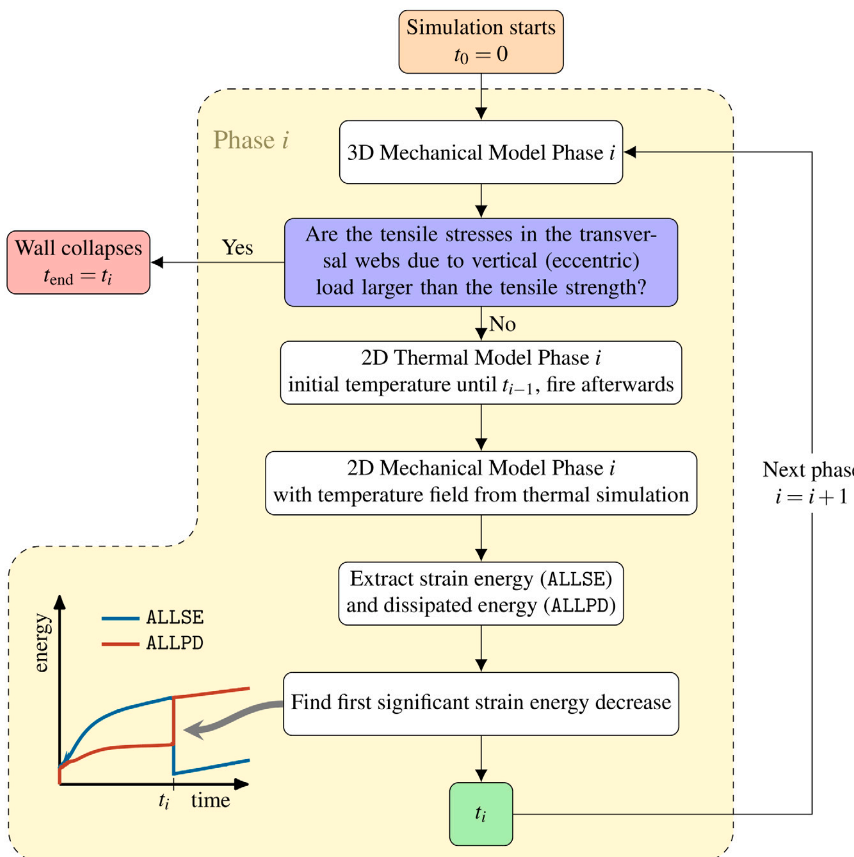


Fig. 15. Flow chart for the entire simulation process explaining the spalling and failure criterion. Spalling of a longitudinal web was predicted considering the evolution of the strain energy and the dissipated energy due to plastic deformations in the 2D mechanical simulation. The obtained spalling time was then used for the 2D thermal simulation of the subsequent Phase. The 3D mechanical simulations were used to identify the failure of the wall.

The coefficient of thermal expansion (CTE) of the clay mixture was determined in a dilatometric analysis. It amounts to $\alpha_{T,c} = 8 \times 10^{-6}\,\mathrm{T}^{-1}$ at $T_{\mathrm{ref}} = 21\,°\mathrm{C}$ and is nearly constant until approximately 400 °C; then it starts to increase sharply, with a peak around 575 °C. This sharp increase occurs due to the conversion of quartz from α-quartz to β-quartz (see e. g. Müller et al. [46]). The phenomenon is called *quartz inversion*, and the increase can reach up to 100 %, depending on the amount of quartz in the clay. After the quartz inversion, the CTE decreases again, being lower than before in general (see e. g. [23]). For the simulations, the CTE at 575 °C was chosen to be 70 % higher than the initial value and nearly 50 % lower than the initial value at higher temperatures (see Fig. 16(b)).

Like the stiffness, the tensile and compressive strengths of extruded fired clay are orthotropic, with the largest strengths observed in the extrusion direction [16]. The strengths normal to this direction depend primarily on the orientation of a specific web. In general, the in-plane strengths are larger than the out-of-plane strengths. Again, the strengths had to be defined as isotropic with CDP as plasticity behavior. The larger in-plane strengths, $\sigma_{f,t,c} = 7.5\,\mathrm{MPa}$ and $\sigma_{f,c,c} = 27.6\,\mathrm{MPa}$, were chosen as tensile and compressive strength for the finite-element simulations, given the same reasoning as for Young's modulus. The material strengths were identified in compressive and three-point bending tests and were assumed to be temperature-independent in the simulations, although a temperature-dependency can be observed in fired clay. In

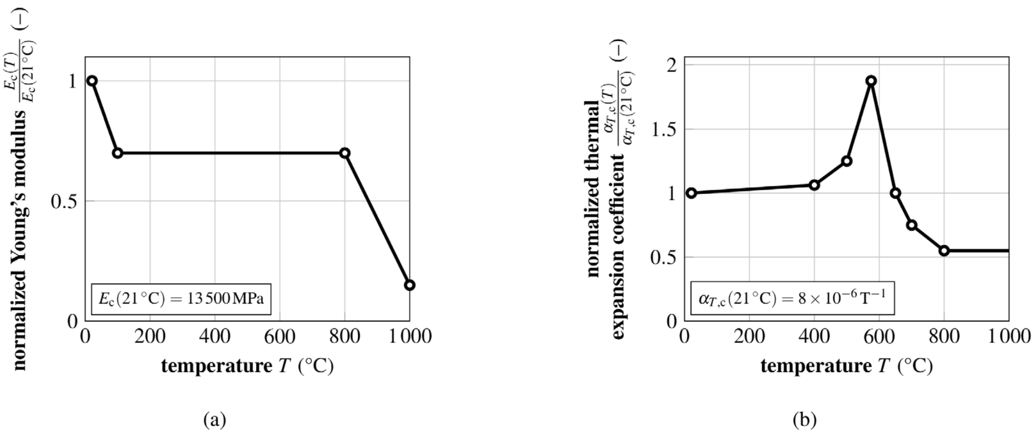


Fig. 16. Mechanical properties of fired clay used in the finite-element model. Young's modulus [11] (a) and the thermal expansion coefficient (b) were considered temperature-dependent.

literature, we found two different contrary evolutions of the material strengths: EN 1996-1-2 [20] describes a linearly decreasing strength with increasing temperature. In contrast, Nguyen and Meftah [11] used an increasing strength with temperature, which they obtained from experiments on different fired clay blocks. A currently running research project revealed a similar trend as Nguyen and Meftah used, corroborating this evolution. Additionally, the decreasing strength evolution is not suggested anymore in the new draft of EN 1996-1-2 in 2022. Therefore, we assume the suggested relation from Nguyen and Meftah to be appropriate.

The temperature–strength-relation Nguyen and Meftah [11] used shows a 10 % increase until 400 °C. Only above 400 °C the strength increases more rapidly up to 74 % of its initial value at 21 °C. The largest stresses leading to spalling occurred in the transversal webs behind the longitudinal web exposed to fire. The nodal temperatures in these regions were always below 500 °C (see also Fig. 17 and Appendix). Therefore, the strength increase can be expected to be around 10 %. This inaccuracy was acceptable in our study, considering the strength variations of fired clay due to defects and the heterogeneous nature of fired clay. Therefore, the simplification of the strength evolution as temperature-independent seems to be appropriate in this case. Thermal creep effects were not considered, given the brittle material behavior and the short duration until failure occurs.

3.4.5. Numerical aspects

The rapidly increasing temperatures combined with the very brittle material behavior led to a numerical problem of high complexity. Therefore, the *Step stabilization* in Abaqus was used to overcome numerical instabilities. This stabilization method introduces an artificial damping factor, which is used to calculate nodal damping forces for nodes with spontaneously rising nodal velocities (see also Simulia [33]).

3.5. Three-dimensional mechanical finite-element model

A three-dimensional model was used to model the stress field due to the vertical loading and to identify the overall failure of the wall. This model is based on the research of Kiefer et al. [16]. In contrast to the two-dimensional models, the mortar joints were also considered. Since the thickness of the thin-bed mortar joint was less than 1 % of the block height, a simplified micro-modeling approach was considered, i. e. mortar was modeled as a 2D cohesive surface connecting the bricks rather than a 3D region (see Lourenço [47]). The models consist of 272686 to 433524 eight-node linear brick elements (C3D8). Analogous to the 2D model, a unit-cell concept with periodic boundary conditions was adopted to minimize the numerical expenses while maintaining a detailed geometry. With a height of two blocks and a width of one block, the chosen unit-cell is the smallest possible unit-cell without considering point and line symmetries (see Fig. 6(a)).

In the three-dimensional formulation of the periodic boundary conditions, not only the *East* and *West* surfaces of the continuum are coupled, but also the top and bottom surfaces. Further details on the implementation of the periodic boundary conditions can be found in Kiefer et al. [16] as well as Suda et al. [29].

The vertical loading of the wall was introduced as vertical stress $\sigma_{zz}(y)$ on the top and bottom surface of the unit-cell. Notably, the progressive spalling of the longitudinal webs leads to a progressively increasing load eccentricity e (see Fig. 6(b)), yielding a linear stress distribution in the thickness direction of the wall, with larger vertical compression on the unexposed side and smaller vertical compression on the fire-exposed side. Thus, these phase-specific vertical stresses $\sigma_{zz}(y)$ read as

$$\sigma_{zz}(y) = \frac{N}{A} + \frac{N \cdot e}{I_x} \cdot y, \tag{18}$$

with the constant vertical load, N, the net cross-sectional area A, the moment of inertia I_x, and the coordinate in thickness direction y, which is related to the phase-specific center of mass of the cross-section.

The elastic properties of fired clay are described in Section 3.4.2, the elastic properties of mortar were overtaken from previous studies [16,29], considering findings from Domone and Illston [48] as well as Sarhosis and Sheng [49]. This way, the orthogonal and tangential interface stiffness for the cohesive surface properties amount to $K_{nn} = 5000\,N/mm$ and $K_{ss} = K_{tt} = 2083\,N/mm$, respectively. Mortar failure was not considered relevant.

4. Model results, validation, and discussion

4.1. Evolution of temperature fields

Fig. 17(a) shows the distribution of nodal temperatures in Phase 4 as a qualitative example for all the modeling phases. Similar graphs for the other five modeling phases are given in the supplementary material. The temperature is around 850 °C at the fire-exposed surface and decreases significantly to approximately 530 °C on the opposite surface of the first longitudinal web. Interestingly, the surface temperature of the second longitudinal web is higher at the center of the cavities compared to the regions close to transversal webs, demonstrating that the heat transfer resistance through the cavity is smaller than the one through the fired clay. Already in the third longitudinal web, the temperature stayed nearly constant. In the cavities, the temperature decrease is less significant, suggesting a smaller heat transfer resistance there. The wavy appearance of the temperature distribution underlines

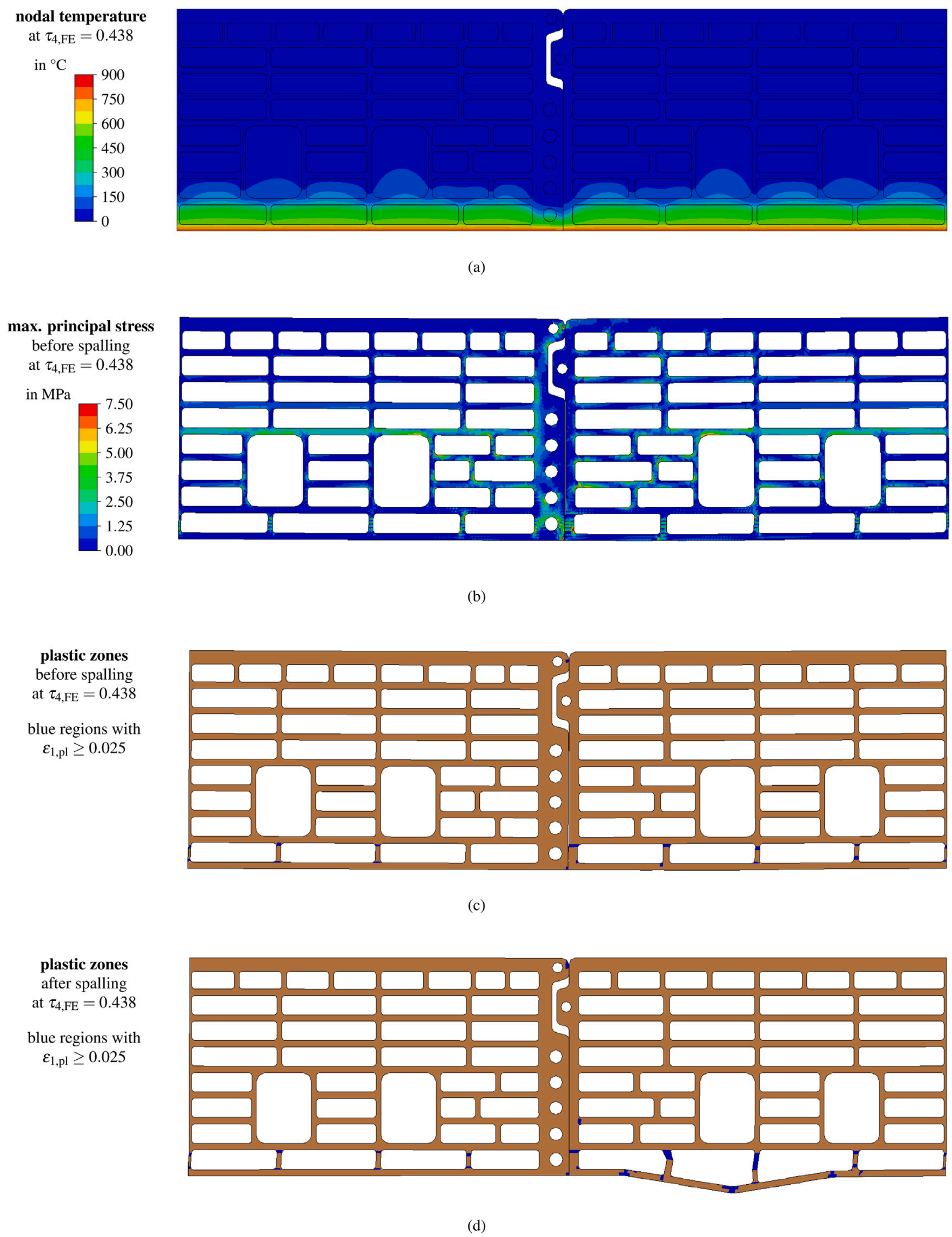


Fig. 17. Nodal temperature (a) and maximum principal stresses (b) in Phase 4 right before estimated spalling time $\tau_{4,\mathrm{FE}} = 0.438$. Plastic zones indicated by the maximum principal plastic strain $\varepsilon_{1,\mathrm{pl}} \geq 0.025$ (blue regions) right before (c) and after (d) this spalling time. Similar graphics for the other five modeling phases are provided in the supplementary material. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

this suggestion since the higher temperatures advanced slightly further through the wall directly behind the air-filled cavities. At the moment of spalling, the large temperature on the *South* side of the model did only advance to the second row of cavities. The nodal temperature stayed near the initial value of 21 °C from the third longitudinal web upwards.

The simulated temperatures were compared to the experimentally measured temperatures inside the cavities to validate the thermal model (see Fig. 18). Notably, the spalling times required for differentiating the six phases were not yet calculated from the mechanical simulations but approximated from the experimental values, as described in Section 2. In general, the thermal simulations agree well with the experiments. The model underestimates the cavity temperature at the beginning of each phase but slightly overestimates it afterward. Notably, the model also nicely predicts the slowdown of the temperature increase related to the evaporation of pore water

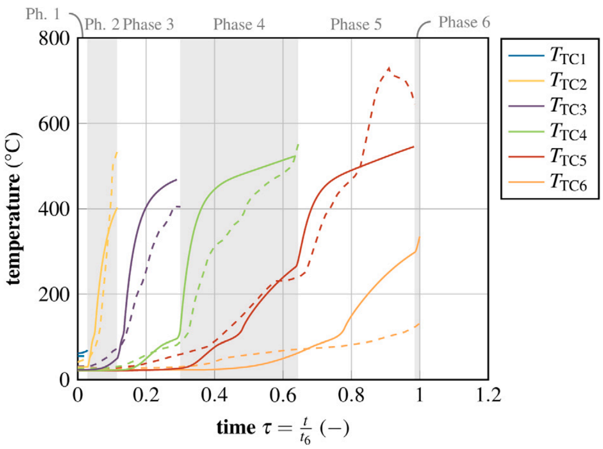


Fig. 18. Comparison of measured (dashed lines) and predicted (solid lines) evolution of cavity temperatures.

observed in almost all phases at around 100 °C. In the sixth cavity, the temperature increase in the simulations is significantly larger than in the experiment. One possible reason for that can be found in the fourth longitudinal web. The spalling of the fourth web was approximated using the temperature increase in the fifth cavity. However, spalling is not the only possible reason for a temperature increase. The fourth web could have been cracked open at another wall location. Then the temperature in the cavity behind might already increase, although the web is still intact. Since the temperature trend differs significantly from the other curves, the thermocouples might also have been corrupted. Either way, the temperatures in the sixth cavity were not that important for the simulation since the last phase of the experiment did not last very long.

The good agreement of simulated and measured temperature evolutions during Phases 1 to 5 underlines the validity of the chosen temperature model and motivates the discussion of the heat transfer mechanism. At the beginning of the simulation, the maximum heat flux could be observed in the transversal webs, while the heat flux in the cavities was low in comparison (see Fig. 19(a)). Around 250 °C, the distribution started to shift, and in the end, nearly all the heat was transferred through the cavities (see Fig. 19(b)). This observation suggests the following conclusion: While the cavities are essential for the high thermal resistance of the block at ambient temperatures, the drastically increasing part of radiant heat transfer in the cavities at higher temperatures has a negative impact on the insulation condition (I according to the classification standard EN 13501-2 [2]).

4.2. Evolution of stress fields and sequential spalling

Fig. 17(b) shows the distribution of the maximum principal stresses in Phase 4 as a qualitative example for all the modeling phases. The most significant tensile stresses occur in the connection between the transversal and longitudinal webs (red areas). Since the outermost longitudinal web expands towards the head joint, the transversal webs closest to the head joint are the most critical. In Phase 4, the fifth longitudinal web shows tensile stresses, which are increasing to the middle of the block. In the earlier phases, such tensile stresses occur in the second longitudinal web; see the supplementary material for corresponding graphs. Due to the large cavities in the middle of the block, these tensile stresses are redistributed to the first longitudinal web behind these large cavities. Compressive stresses occur in the outermost longitudinal web, counteracting these tensile stresses. Note that these compressive stresses cannot be seen in Fig. 17(b), due to the choice of the shown variable.

Besides the negative impact on the insulation condition, the decrease in effective thermal resistance at higher temperatures (shown in Fig. 19) also has a positive effect on the performance of the wall in the fire experiment: Large temperature gradients are the reason for large stresses in the block, which lead to failure of the webs (this affects the parameters R and E in the classification standard EN 13501-2 [2]). A loss in the overall thermal resistance yields a faster decrease of these temperature gradients and, therefore, smaller stresses.

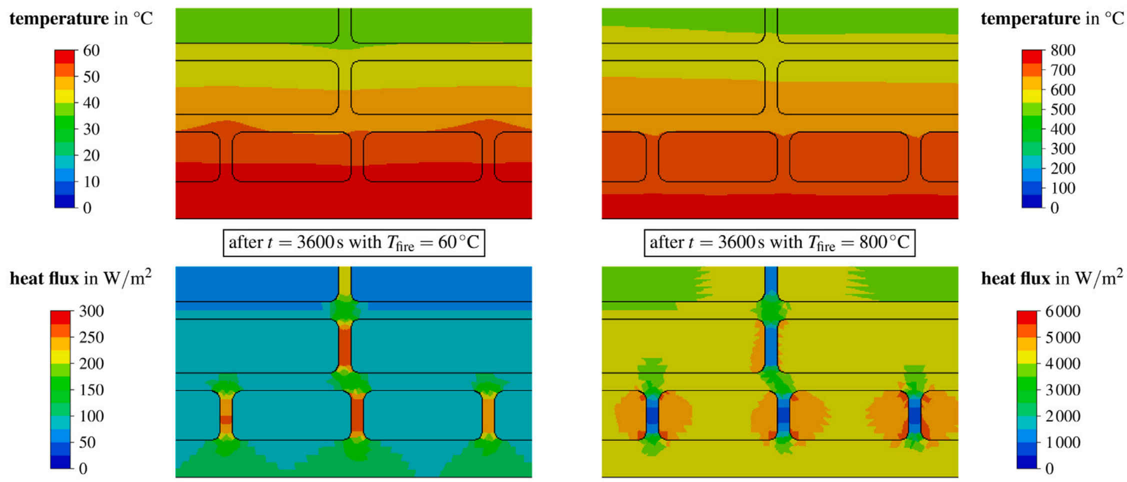
Next, the identification of the spalling times is discussed. The plastic deformations mainly occur in the connection between the transversal and longitudinal webs. Nevertheless, the plastic dissipated energy in these regions dominates the dissipated energy of the whole model. Thus, the evolution of the total energy, strain energy, and dissipated energy of the whole 2D model are plotted in Fig. 20 for each of the six model phases. Notably, the time values $\tau^{\mathrm{ph},i}$ on the horizontal axis are related to the start of the corresponding phase rather than the failure of the insulation, as the total time τ is. Interestingly, the total energy curves of Phases 1, 4, and 5 increase concavely, which are the models where the first longitudinal webs are continuous and in contact in the head joint (see the block geometry depicted in Fig. 2). In both phases, where the outermost webs are not in contact at the head joint (Phases 2 and 3), the strain energy decreases before it increases concavely. After some time, the dissipated energy increases abruptly (revealed by the jump of the red line in Fig. 20) while the strain energy decreases in the same way, at least for Phases 1 to 4. This time was considered the model-predicted spalling time. The deformations and plastic strains in the model changed at these moments (e. g. Figs. 17(c) and 17(d) for Phase 4): While just a few transversal webs between the two longitudinal partitions closest to the fire showed plastic zones before the derived time, each transversal web in this row showed significant plastic strains afterward. Additionally, the longitudinal web closest to the fire buckled, as already demonstrated by Nguyen and Meftah [13]. Most notably, the experimentally measured spalling time for Phases 1 to 4 is remarkably close to the predicted counterpart (see Fig. 20), which corroborates the two-dimensional mechanical model and its underlying assumptions.

While the plastic zones in Phases 1 to 4 concentrate in the outermost transversal webs, they are much more evenly distributed over the wall thickness in Phases 5 and 6. This observation can be traced back to the large holes in the middle of the brick (see Fig. 2). The web structure is less rigid than in the phases before due to the non-continuous longitudinal webs six and seven and the offset transversal webs. Thus, temperature-induced deformations are less hindered, leading, in turn, to smaller stresses and a less brittle behavior without pronounced jumps in the evolution of the cumulative dissipated energy. Nevertheless, in Phase 5, two small jumps could be observed (see Fig. 20(e)). The range between the corresponding time instants ($\tau^{\mathrm{ph},5} = 0.196$ and $\tau^{\mathrm{ph},5} = 0.587$) was considered as a possible interval where the longitudinal web spalled. This interval nicely bounds the experimentally determined spalling time amounting to 0.341. In Phase 6, a small dissipation burst occurs at $\tau^{\mathrm{ph},6} = 0.188$, but this time instant is already well after the experimentally determined collapse of the wall. This way, we expect the stresses due to the already significant eccentric vertical loading to be critical, as analyzed next.

4.3. Stresses due to vertical loading and ultimate failure of the wall

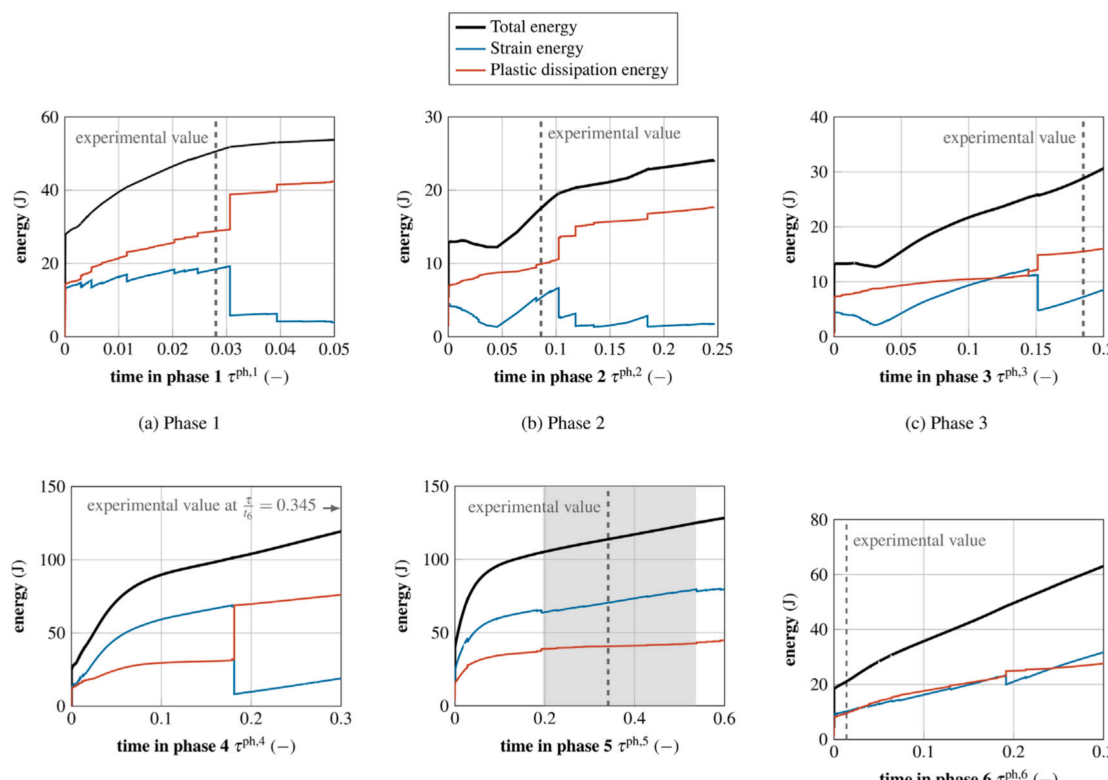
Tensile stresses in the transversal webs lead to the failure of vertically perforated clay block masonry under vertical compression (see Suda et al. [29]). Since the study focuses on the spalling of the outermost longitudinal webs, the maximum tensile stresses in the transversal webs right behind the first longitudinal web are the most interesting to observe (see Fig. 21). These stresses are large in the bed joints and rapidly decrease when going vertically to the middle of the block.

In the first five phases, the stresses are below the tensile strength (see Table 3). Therefore, the vertical loading alone would not lead



(a) (b)

Fig. 19. Detail of temperature (top) and heat flux (bottom) in the first three longitudinal webs at (a) $t = 3600$ s with firing temperature (at the bottom surface) of 60 °C, showing that heat is transferred preferably through transversal webs, and (b) $t = 3600$ s with firing temperature 800 °C, showing that heat is transferred preferably through cavities.



(a) Phase 1 (b) Phase 2 (c) Phase 3

(d) Phase 4 (time according to experiment is 240 s) (e) Phase 5. The filled area indicates the possible failure range. (f) Phase 6

Fig. 20. Whole model energy values. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Ratio of the largest tensile stresses to the material's tensile strength in the outermost transversal webs of the corresponding model phase.

Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6
21.3 %	37.6 %	49.2 %	73.6 %	95.3 %	199.7 %

to the failure of the wall in these phases. However, the tensile stress in Phase 5 is only insignificantly below the tensile strength. From Phase 5 to Phase 6, the maximum tensile stress more than doubles and is well above the tensile strength. The underlying reason for this increase is the outermost longitudinal web being non-continuous in Phase 6. Therefore, we consider that the wall collapses at the end of Phase 5 when the fifth longitudinal web spalls. The observations in the experiment support this modeling result since the wall endured only a few seconds after Phase 6 was reached.

Finally, the time until collapse predicted by the model is compared to the experimentally measured collapse time. Fig. 22 shows

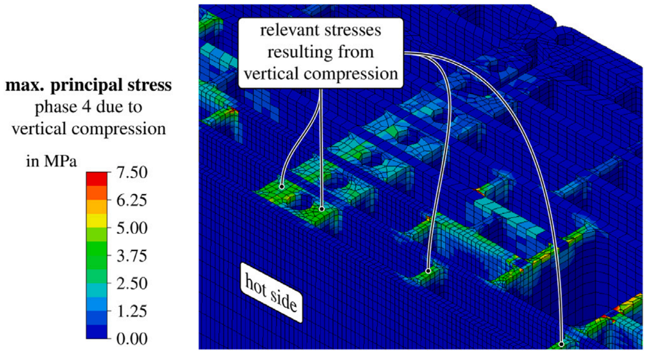


Fig. 21. Maximum principal stresses in the transversal webs of the three-dimensional mechanical model in Phase 4 due to vertical loading.

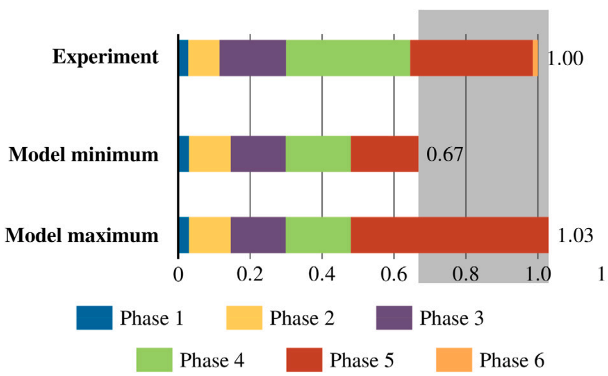


Fig. 22. Comparison of model-predicted and experimentally determined time until the collapse of the wall, subdivided into the six phases corresponding to the sequential spalling of the outermost webs; the gray area in the background marks the interval spanned by the two limit cases predicted by the model.

the overall time of the simulations compared to the experiment. Given the predicted spalling range in Phase 5 (Fig. 20(e)), two bounds were predicted for the total time until collapse, see Fig. 22. The experimentally measured collapse time nicely falls in between these two bounds. It is reasonable to be closer to the upper bound because of the implemented unit-cell concept: The same failure state is assumed over the whole width of the wall since the unit-cell is repeated periodically. The specimen in the experiment started to fail in the center of the wall, while other regions stayed intact. Therefore, a shorter overall time in the simulations seems reasonable.

5. Conclusion

Within this work, a coupled temperature–displacement finite-element model for assessing the performance of vertically perforated clay block masonry in a firing test was presented and compared to novel experimental data. Using an energy-based spalling criterion combined with a unit-cell approach allowed for modeling only a small region and, thus, saved numerical expenses. Hence, non-linear and temperature-dependent material behavior and fine meshing could be used, enabling detailed insights into the distribution and evolution of stresses, strains, and local damage zones. Since the model only considers a small region of the whole wall, no information about the overall deformation could be gained, such as obtained in e. g. Nguyen and Meftah [11]. Nevertheless, the presented approach contributes to a better understanding of the complex mechanisms involved in the behavior of vertically perforated clay block masonry in fire situations.

The simulation was split into six sequential phases with slightly different geometry, considering spalling of the longitudinal webs as the critical mechanism for failure. Spalling of the webs could be decoupled from the vertical loading; thus, a two-dimensional modeling approach was sufficient for simulating the firing test. Nonetheless, purely elastic three-dimensional mechanical models helped interpret the results from the two-dimensional models and depict the failure of the entire wall.

The temperatures simulated with the proposed two-dimensional thermal model show good agreement with the experiments, even after spalling of the longitudinal webs. Additionally, the thermal simulations revealed a redistribution of the heat flow from the transversal webs to the air-filled cavities at higher temperatures. This change in heat transfer is driven by the increasing impact of radiation and convection at higher temperatures. Therewith, a positive and negative effect of the air-filled cavities on the performance in the firing test could be derived.

A novel energy-based spalling criterion was presented for predicting the longitudinal webs' spalling. By identifying a qualitative decrease in strain energy (and vice versa, an increase of dissipated energy), this criterion allowed the depiction of spalling for each phase.

Thus, considering the temperatures of the two-dimensional thermal models, the energy evolution of the two-dimensional mechanical models, and the maximum principal stresses of the three-dimensional mechanical models for each phase, we were able to predict the performance of a vertically perforated clay block wall in a firing test. The simulated performance showed good agreement with experimental measurements, notably without any empirical fitting parameters.

In general, finding an optimum for vertically perforated clay blocks is a hard-to-achieve goal since there are masses of existing product designs and infinite possible new configurations. Performing a parameter study for a specific design seems to be a more reasonable approach. Hence, utilizing the available computational methods by extending the block design experience by numerical approaches like the presented model could help to improve the block design process.

CRediT authorship contribution statement

Raphael Reismüller: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Visualization. **Markus Königsberger:** Conceptualization, Writing – review & editing. **Andreas Jäger:** Conceptualization, Supervision, Project administration, Funding acquisition. **Josef Füssl:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Data availability

The data that has been used is confidential.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.firesaf.2022.103729.

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